

Question Generation

Template

Had **{event}** occurred at time **{t}** ms, what would be the expected values of **{joint_signal}** at $T + \{n\}$ ms in this counterfactual case?

{event} $\leftarrow$ +0.5 kg payload
{t} $\leftarrow$ 4 329
{joint_signal} $\leftarrow$ joint accel.
{n} $\leftarrow$ 500

Instantiated Question

"Had an increase of 0.5 kg to the payload occurred at time 4 329 ms, what would be the expected values of joint accelerations at $T + 500$ ms in this counterfactual case?"

Ground Truth: Answer Generation

Deterministic Answer Rules

Algorithm (Template L3-5, Tensor):

1. Load counterfactual episode (fault-injected)
2. Find row at timestamp $t + n$ ms
3. Extract joint_accel_0 ... joint_accel_5
4. Ground truth = $[v_0, v_1, v_2, v_3, v_4, v_5]$

Other L3 formats: Multi-Select, Ranking

episode data



evaluate against data

Ground-Truth Answer

Format: Tensor [6]

Answer: **1.23_0.87_-0.42_0.15_2.01_-0.03**

Also generated for Level 3:

Multi-Select (T/F string) — 2 templates

Ranking (permutation ABCD) — 1 template